

MMS source terms for the `logc_muh` + multi-ion + Stern stack

PNPInverse / Forward solver verification

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Abstract

A brief derivation of the four manufactured-solution source terms required to convergence-test the production PNP-BV stack used by `scripts/studies/solver_demo_slide15_no_speculative_cs.py`: muh-formulation proton primary variable, three dynamic species $\{\text{O}_2, \text{H}_2\text{O}_2, \text{H}^+\}$, two analytic Bikerman counterions $\{\text{Cs}^+, \text{SO}_4^{2-}\}$ in the shared- θ closure, Stern Robin BC at the electrode, and parallel R2e/R4e Butler-Volmer in log-rate form. Full discussion (clip-safety bounds, runtime asserts, coverage table) is in `docs/solver/mms_pnpbv_muh_multi_ion_stern_derivation.md`; this writeup captures only the algebra of the source injections.

1 Setup

Primary unknowns. On a mixed FE space $W = V_{\text{O}_2} \times V_{\text{H}_2\text{O}_2} \times V_{\mu_H} \times V_\varphi$:

$$U = (u_{\text{O}_2}, u_{\text{H}_2\text{O}_2}, \mu_H, \varphi), \quad z = (0, 0, +1, \cdot), \quad e_m z_H = +1.$$

The proton enters via its electrochemical potential $\mu_H = u_H + e_m z_H \varphi$; concentrations are reconstructed as $c_H = \exp(\mu_H - e_m z_H \varphi)$, $c_i = \exp(u_i)$ for $i \in \{\text{O}_2, \text{H}_2\text{O}_2\}$.

Domain and BCs. $\Omega = (0, 1)^2$ with electrode marker at $y = 0$, bulk at $y = 1$. Bulk Dirichlets $u_i = \ln c_{0,i}$, $\mu_H = \ln c_{0,H}$, $\varphi = 0$. Electrode: BV-flux Neumann on c_i (implicit) and a Stern Robin on φ . Side walls: natural zero-flux ($J_i \cdot n = 0$ for species, $\nabla \varphi \cdot n = 0$ for Poisson).

Shared- θ multi-ion Bikerman closure. For counterions $k \in \{\text{Cs}^+, \text{SO}_4^{2-}\}$ with valence z_k , bulk concentration c_k^{bulk} , and hard-sphere a_k^{ctn} ,

$$q_k(\varphi) = \exp(-z_k \varphi), \tag{1}$$

$$A_{\text{dyn}}(c_{\text{dyn}}) = \sum_{i \in \text{dyn}} a_i^{\text{dyn}} c_i, \tag{2}$$

$$\theta_b = 1 - \sum_{i \in \text{dyn}} a_i^{\text{dyn}} c_{0,i} - \sum_k a_k^{\text{ctn}} c_k^{\text{bulk}}, \tag{3}$$

$$D(\varphi) = \theta_b + \sum_k a_k^{\text{ctn}} c_k^{\text{bulk}} q_k(\varphi), \tag{4}$$

$$c_k^{\text{ster}}(\varphi; c_{\text{dyn}}) = c_k^{\text{bulk}} q_k(\varphi) \frac{1 - A_{\text{dyn}}}{D(\varphi)}, \tag{5}$$

$$\rho_k(\varphi; c_{\text{dyn}}) = z_k c_k^{\text{ster}}, \quad P_k(\varphi; c_{\text{dyn}}) = a_k^{\text{ctn}} c_k^{\text{ster}}. \tag{6}$$

The steric chemical potential acting on dynamic species:

$$\theta_{\text{inner}}(\varphi; c_{\text{dyn}}) = 1 - A_{\text{dyn}} - \sum_k P_k, \quad \mu_{\text{ster}} = -\ln \theta_{\text{inner}}. \quad (7)$$

(Per-ion φ -clamps, `free_dyn_floor`, and `packing_floor` are identity at the manufactured solution by design; see the markdown writeup §5.)

Production residual. Production builds the residual (mobility-flux convention $J_i = +D_i c_i \nabla(\cdot)$, not the physical $N_i = -J_i$):

$$\begin{aligned} F_{\text{res}} \ni & \sum_i \int_{\Omega} \frac{c_i - c_{i,\text{old}}}{\Delta t} v_i \, dx + \sum_i \int_{\Omega} J_i \cdot \nabla v_i \, dx - \sum_{i,j} \int_{\Gamma_{\text{elec}}} s_{ij} R_j v_i \, ds \\ & + \int_{\Omega} \varepsilon_c \nabla \varphi \cdot \nabla w \, dx - \int_{\Omega} \rho_c \left(\sum_i z_i c_i + \sum_k z_k c_k^{\text{ster}} \right) w \, dx - \int_{\Gamma_{\text{elec}}} C_S^{\text{model}} (\varphi_{\text{app}}^{\text{model}} - \varphi) w \, ds, \end{aligned} \quad (8)$$

with $\varepsilon_c = \text{scaling}[\text{'poisson_coefficient'}]$, $\rho_c = \text{scaling}[\text{'charge_rhs_prefactor'}]$, $C_S^{\text{model}} = \text{scaling}[\text{'bv_stern_capacitance_model'}]$, and $\varphi_{\text{app}}^{\text{model}} = V_{\text{RHE}}/V_T$ (nondim).

Butler–Volmer log-rates. For $j \in \{R_{2e}, R_{4e}\}$ in the Stern-on branch (η_j depends on the local φ):

$$\eta_j(x, y) = \text{bv_exp_scale}(\varphi_{\text{app}}^{\text{model}} - \varphi(x, y) - E_{eq,j}^{\text{model}}), \quad (9)$$

$$\log r_{j,\text{cath}} = \ln k_{0,j} + u_{c_j} + \sum_f p_f (u_{s_f} - \ln c_{\text{ref},f}) - \alpha_j n_{e,j} \eta_j, \quad (10)$$

$$\log r_{j,\text{anod}} = \begin{cases} \ln k_{0,j} + u_{a_j} + (1 - \alpha_j) n_{e,j} \eta_j & (\text{reversible, anodic species defined}) \\ -\infty & (\text{R4e: irreversible} \Rightarrow \text{anodic} = 0) \end{cases} \quad (11)$$

$$R_j = \exp(\log r_{j,\text{cath}}) - \exp(\log r_{j,\text{anod}}). \quad (12)$$

Critical muh substitution: wherever u_H appears in the BV expressions, the muh-aware code uses $u_H^{\text{recon}} = \mu_H - e_m z_H \varphi$, not the raw μ_H primary.

2 Manufactured solution

Choose smooth fields satisfying all bulk Dirichlets exactly:

$$c_i^{\text{ex}}(x, y) = c_{0,i} [1 + \delta_i \cos(\pi x) (1 - y)^2], \quad i \in \{O_2, H_2O_2, H\}, \quad (13)$$

$$\varphi^{\text{ex}}(x, y) = (1 - y) [\alpha_0 + \alpha_1 \cos(\pi x)] + \gamma y (1 - y) \cos(\pi x), \quad (14)$$

$$\mu_H^{\text{ex}}(x, y) = \ln c_H^{\text{ex}}(x, y) + e_m z_H \varphi^{\text{ex}}(x, y), \quad (15)$$

with envelope $\delta_i = 0.30$, $\alpha_0 = \alpha_1 = \gamma = 0.5$. At $y = 1$, $c_i^{\text{ex}} = c_{0,i}$ and $\varphi^{\text{ex}} = 0$ (bulk Dirichlets met). At $y = 0$, $\varphi^{\text{ex}}(x, 0) = \alpha_0 + \alpha_1 \cos(\pi x) \in [0, 1]$, which differs from $\varphi_{\text{app}}^{\text{model}} \approx 21.41$ so the Stern Robin is non-trivially exercised. At $x \in \{0, 1\}$, $\partial_x = 0$ for every field, so side-wall natural BCs are auto-satisfied and no side-wall source is needed.

Reconstructed exact closure quantities are obtained by direct UFL composition at the manufactured fields:

$$A_{\text{dyn}}^{\text{ex}} = \sum_i a_i^{\text{dyn}} c_i^{\text{ex}}, \quad D^{\text{ex}} = \theta_b + \sum_k a_k^{\text{ctn}} c_k^{\text{bulk}} q_k^{\text{ex}}, \quad (16)$$

$$c_k^{\text{ster,ex}} = c_k^{\text{bulk}} q_k^{\text{ex}} \frac{1 - A_{\text{dyn}}^{\text{ex}}}{D^{\text{ex}}}, \quad \theta_{\text{inner}}^{\text{ex}} = 1 - A_{\text{dyn}}^{\text{ex}} - \sum_k P_k^{\text{ex}}, \quad (17)$$

$$\mu_{\text{ster}}^{\text{ex}} = -\ln \theta_{\text{inner}}^{\text{ex}}, \quad (18)$$

$$J_i^{\text{ex}} = D_i c_i^{\text{ex}} (\nabla u_i^{\text{ex}} + \nabla \mu_{\text{ster}}^{\text{ex}}), \quad i \in \{O_2, H_2O_2\}, \quad (19)$$

$$J_H^{\text{ex}} = D_H c_H^{\text{ex}} (\nabla \mu_H^{\text{ex}} + \nabla \mu_{\text{ster}}^{\text{ex}}). \quad (20)$$

3 Source terms

Pattern. For each production weak-form contribution $+\int X v \, d\mu$, subtract a source $-\int X|_{u_{\text{ex}}} v \, d\mu$ so that the discrete operator drives U_h towards the manufactured field at the FE discretization rate. We do not use `ufl.replace` on F_{res} itself: each source integrand is written from the strong-form residual, independent of the discrete weak form, so that any wiring bug in F_{res} shows up as a convergence-rate violation rather than canceling symmetrically.

3.1 Nernst–Planck interior source

Integrating $\int J_i \cdot \nabla v_i \, dx$ by parts moves the divergence onto J_i and produces a boundary trace on $\partial\Omega$. The interior strong-form residual at u_{ex} is $-\text{div } J_i^{\text{ex}}$, so the interior NP source is

$$\boxed{S_{c_i}(x, y) = -\text{div } J_i^{\text{ex}}(x, y), \quad F_{\text{res}} \leftarrow F_{\text{res}} - \int_{\Omega} S_{c_i} v_i \, dx,} \quad (21)$$

implemented as `fd.div(J_i_ex)` with elevated quadrature `SRC_QUAD_DEGREE` ≥ 8 (final value pinned by a one-time sweep on $N = 32$). The boundary trace $\int J_i^{\text{ex}} \cdot n v_i \, ds$ vanishes on bulk (Dirichlet test functions) and on side walls ($\partial_x c_i^{\text{ex}} = 0$); the electrode trace is handled below.

3.2 Nernst–Planck electrode boundary source

At $y = 0$ the production residual contains, after IBP, an implicit $\int J_i \cdot n v_i \, ds$ from the Laplacian plus an explicit $-\int \sum_j s_{ij} R_j v_i \, ds$ from the BV term. The natural BC at u_{ex} is therefore $J_i^{\text{ex}} \cdot n = \sum_j s_{ij} R_j^{\text{ex}}$; the gap between the manufactured profile and this natural BC is supplied as a source:

$$\boxed{g_i^{\text{elec}}(x) = J_i^{\text{ex}} \cdot n|_{y=0} - \sum_j s_{ij} R_j^{\text{ex}}|_{y=0}, \quad F_{\text{res}} \leftarrow F_{\text{res}} - \int_{\Gamma_{\text{elec}}} g_i^{\text{elec}} v_i \, ds.} \quad (22)$$

The outward normal at $y = 0$ is $n = (0, -1)$, so for the mobility-convention fluxes

$$J_i^{\text{ex}} \cdot n|_{y=0} = -D_i c_i^{\text{ex}}|_{y=0} \partial_y (u_i^{\text{ex}} + \mu_{\text{ster}}^{\text{ex}})|_{y=0}, \quad i \in \{O_2, H_2O_2\}, \quad (23)$$

$$J_H^{\text{ex}} \cdot n|_{y=0} = -D_H c_H^{\text{ex}}|_{y=0} \partial_y (\mu_H^{\text{ex}} + \mu_{\text{ster}}^{\text{ex}})|_{y=0}. \quad (24)$$

Implementations should compose J_i^{ex} as a UFL expression and use `fd.dot(J_i_ex, fd.FacetNormal(mesh))` rather than expanding the sign by hand. $R_j^{\text{ex}}|_{y=0}$ is genuinely x -dependent in the Stern-on branch because η_j^{ex} varies with $\varphi^{\text{ex}}(x, 0)$.

Sign sanity. For O_2 with $s = -1$ and cathodic $R_{R2e} > 0$, the natural BC gives $J_{O_2}^{\text{ex}} \cdot n = -R_{R2e} < 0$ at $y = 0$, hence $J_{O_2,y}^{\text{ex}} > 0$ and $\partial_y u_{O_2}^{\text{ex}} > 0$: c_{O_2} increases away from the electrode (boundary-layer depletion), as expected.

3.3 Poisson interior source

IBP on the Laplacian: $\int \varepsilon_c \nabla \varphi \cdot \nabla w \, dx = - \int \varepsilon_c \Delta \varphi w \, dx + \int \varepsilon_c \nabla \varphi \cdot n w \, ds$. The interior strong-form residual at u_{ex} collects the Laplacian piece with both charge densities:

$$S_\varphi(x, y) = -\varepsilon_c \Delta \varphi^{\text{ex}} - \rho_c (z_H c_H^{\text{ex}}) - \rho_c \sum_k z_k c_k^{\text{ster,ex}}, \quad F_{\text{res}} \leftarrow F_{\text{res}} - \int_\Omega S_\varphi w \, dx_q. \quad (25)$$

For the chosen φ^{ex} ,

$$\Delta \varphi^{\text{ex}} = -\pi^2 \cos(\pi x) [(1-y)\alpha_1 + \gamma y(1-y)] - 2\gamma \cos(\pi x). \quad (26)$$

Implementations use `fd.div(fd.grad(phi_ex))` directly. The bulk trace vanishes (Dirichlet); the side-wall trace vanishes ($\partial_x \varphi^{\text{ex}} = 0$ at $x \in \{0, 1\}$); the electrode trace is the Stern Robin source below.

3.4 Stern Robin boundary source

At $y = 0$, the φ -equation residual aggregates the IBP byproduct $\int \varepsilon_c \nabla \varphi \cdot n w \, ds$ from the Laplacian and the explicit Stern term $-\int C_S^{\text{model}}(\varphi_{\text{app}}^{\text{model}} - \varphi) w \, ds$. Subtracting their value at u_{ex} :

$$g_S(x) = \varepsilon_c (\nabla \varphi^{\text{ex}} \cdot n)|_{y=0} - C_S^{\text{model}}(\varphi_{\text{app}}^{\text{model}} - \varphi^{\text{ex}}|_{y=0}), \quad F_{\text{res}} \leftarrow F_{\text{res}} - \int_{\Gamma_{\text{elec}}} g_S w \, ds_q. \quad (27)$$

With $n = (0, -1)$ at $y = 0$,

$$(\nabla \varphi^{\text{ex}} \cdot n)|_{y=0} = -\partial_y \varphi^{\text{ex}}|_{y=0} = \alpha_0 + (\alpha_1 - \gamma) \cos(\pi x), \quad (28)$$

$$\varphi^{\text{ex}}|_{y=0} = \alpha_0 + \alpha_1 \cos(\pi x), \quad (29)$$

so

$$g_S(x) = \varepsilon_c [\alpha_0 + (\alpha_1 - \gamma) \cos(\pi x)] - C_S^{\text{model}}[\varphi_{\text{app}}^{\text{model}} - \alpha_0 - \alpha_1 \cos(\pi x)]. \quad (30)$$

The sign on the $-C_S^{\text{model}}(\cdot)$ piece matches production (`forms_logc_muh.py:668` has $F_{\text{res}} -= C_S^{\text{model}}(\varphi_{\text{app}} - \varphi) w \, ds$), so subtracting $g_S \cdot w \, ds$ from F_{res} exactly cancels the residual at u_{ex} .

4 Summary of what the four source terms catch

Source piece	Catches a bug in
$S_{c_i}, i \in \{O_2, H_2O_2\}$	μ_{ster} inclusion in NP flux; shared- θ $\text{Cs}^+/\text{SO}_4^{2-}$ contributions to packing
S_{c_H}	muh reconstruction ($\nabla u_H \rightarrow \nabla \mu_H$); μ_{ster} coupling for the proton
g_i^{elec} with muh-aware R_j^{ex}	muh substitution $u_H \rightarrow \mu_H - e_m z_H \varphi$ inside BV cathodic factors (powers 2 vs 4); α_1
spatial $\eta_j^{\text{ex}}(x, 0)$	Stern-on branch of <code>_build_eta_clipped</code> ; φ -coupling into BV that only fires when C
S_φ interior	shared- θ numerator ($1 - A_{\text{dyn}}$); per-ion φ -clamp inside $D(\varphi)$; sign on $\rho_c \sum_k z_k c_k^{\text{ster}}$
g_S	sign of $-C_S^{\text{model}}(\varphi_{\text{app}} - \varphi)$; use of the ctx-stored Stern coefficient in the residual; IB

5 Out of scope

By construction the four sources above do not exercise (i) inactive clamp/floor branches (u_{clamp} , `phi_clampk`, `exponent_clip`, `free_dyn_floor`, `packing_floor` – all identity at u_{ex}); (ii) runtime mutation of ctx values (voltage continuation, `set_stern_capacitance_model`, K_0 ramps, z -scale ramps); (iii) the Stern coefficient’s nondim conversion at form-build time (source and residual both consume the converted value); (iv) the c_{ref} -anchored anodic branch of BV (neither R2e nor R4e enters it); (v) the legacy per-species `bv_c_ref_model_vals` path (gated off by `use_reactions = True`); (vi) IC machinery and continuation/anchor logic; (vii) Phase-6 speculative physics (water self-ionization, cation hydrolysis).

The full coverage table, runtime invariants (boundary-norm bound $10 < \|R_{4e}^{\text{ex}}\|/\|R_{2e}^{\text{ex}}\| < 10^5$ enforcing $K_{0,R4e}^{\text{factor}} = 10^{-18}$, discrete-min check on $\theta_{\text{inner}}^{\text{ex}}$, stack-invariant asserts), and parameter envelope justification are in `docs/solver/mms_pnpbv_muh_multi_ion_stern_derivation.md` (post-GPT critique session 38, verdict APPROVED).

6 Verification results

Implementation lands in `scripts/verification/mms_pnpbv_muh_multi_ion_stern.py` (source builder; ~ 1100 LOC) and `tests/test_mms_logc_muh_multi_ion_stern.py` (18-test suite). All tests pass on Firedrake/PETSc as of 2026-05-14.

Convergence rates. A four-mesh `UnitSquareMesh` sweep on $N \in \{8, 16, 32, 64\}$ with `SRC_QUAD_DEGREE = 8` recovers textbook CG1 convergence on every primary unknown (log-log linear regression on the four-mesh chain; $R^2 > 0.99$ in every row):

Field	L^2 rate	H^1 rate	$L^2 R^2$	$H^1 R^2$
u_{O_2}	1.9974	0.9991	> 0.999	> 0.999
$u_{H_2O_2}$	1.9972	0.9993	> 0.999	> 0.999
μ_H	1.9999	1.0003	> 0.999	> 0.999
φ	2.0086	1.0011	> 0.999	> 0.999

Full per-mesh error table:

N	h	L^2 error				H^1 error			
		u_{O_2}	$u_{H_2O_2}$	μ_H	φ	u_{O_2}	$u_{H_2O_2}$	μ_H	φ
8	0.1250	1.97×10^{-3}	4.13×10^{-3}	4.21×10^{-3}	2.13×10^{-3}	7.03×10^{-2}	7.05×10^{-2}	1.80×10^{-1}	1.34×10^{-1}
16	0.0625	5.01×10^{-4}	1.06×10^{-3}	1.06×10^{-3}	5.15×10^{-4}	3.55×10^{-2}	3.55×10^{-2}	8.96×10^{-2}	6.67×10^{-2}
32	0.0312	1.26×10^{-4}	2.67×10^{-4}	2.64×10^{-4}	1.27×10^{-4}	1.78×10^{-2}	1.78×10^{-2}	4.48×10^{-2}	3.33×10^{-2}
64	0.0156	3.15×10^{-5}	6.70×10^{-5}	6.60×10^{-5}	3.16×10^{-5}	8.89×10^{-3}	8.89×10^{-3}	2.24×10^{-2}	1.66×10^{-2}

Newton converges in 2–3 iterations on every mesh (well below the 30-iteration cap) using a backtracking line search (`snese_line_search_type = bt`, `snese_line_search_maxlambda = 0.5`) and tight tolerances (`snese_atol = snese_rtol =  $10^{-8}$`). The default PETSc 12 line search overshoots on this multi-ion + Stern stack and triggers `DIVERGED_DTOL` on the first Newton step; `bt` resolves it cleanly.

Quadrature degree. A pilot sweep at $N \in \{32, 64\}$ with `SRC_QUAD_DEGREE` $\in \{6, 8, 10, 12, 16\}$ yields field errors that agree to all reported digits for every degree ≥ 8 . `SRC_QUAD_DEGREE` = 8 is therefore pinned for the test suite; degree 6 is marginal and not used.

Graded production mesh recovery. A single-solve check on the demo’s graded rectangle ($N_x = 8$, $N_y = 80$, $\beta = 3$ — identical to the mesh `scripts/studies/solver_demo_slide15_no_speculative_cs.py` uses) verifies that Newton recovers the manufactured solution within the expected $\mathcal{O}(h_x^2)$ floor of the coarse x -direction:

Field	L^2 error	H^1 error
u_{O_2}	1.40×10^{-3}	5.45×10^{-2}
$u_{H_2O_2}$	2.41×10^{-3}	5.46×10^{-2}
μ_H	2.93×10^{-3}	1.57×10^{-1}
φ	1.55×10^{-3}	1.19×10^{-1}

Newton: 3 iterations. Test-suite thresholds are pinned at $\sim 6\times$ above these baseline values to absorb compiler / solver-version drift while remaining $\sim 10^2\times$ below what an $\mathcal{O}(1)$ wiring bug would produce.

Broken-config invariant suite. Twelve parametrized tests verify that the prebuild and postbuild invariant phases catch every known way the test problem can silently drift from the production stack:

Mutation	Layer	Match pattern
<code>formulation = logc</code>	prebuild	<code>formulation</code>
<code>bv_log_rate = False</code>	prebuild	<code>bv_log_rate</code>
<code>enable_water_ionization on</code>	prebuild	<code>water_ionization</code>
<code>enable_cation_hydrolysis on</code>	prebuild	<code>cation hydrolysis</code>
<code>$\Delta t = 0.1$ (time term not negligible)</code>	prebuild	<code>dt</code>
<code>$\text{snes_atol} = 10^{-2}$ (too loose)</code>	prebuild	<code>snes_atol</code>
<code>suppress_poisson_source on</code>	prebuild	<code>suppress_poisson_source</code>
<code>R_{4e} reaction missing</code>	prebuild	<code>rxns</code>
<code>SO_4^{2-} counterion missing</code>	prebuild	<code>bikerman</code>
<code>$\text{Cs}^+ \rightarrow \text{ClO}_4^-$</code>	prebuild	<code>Cs, a_Cs, c_b_Cs</code>
<code>Stern off ($C_S = \text{None}$)</code>	postbuild	<code>bv_stern_capacitance_model</code>
<code>$K_{0,R_{4e}}^{\text{factor}} = 1$</code>	postbuild	<code>k0_R4e</code>

All twelve raise `AssertionError` at the documented invariant layer with a recognizable message that `pytest.raises(match=...)` catches.

Acceptance. The full 18-test suite (`TestMMSConvergence`: 3 tests; `TestMMSProductionGradedMesh`: 3 tests; `TestMMSAsserts`: 12 tests) passes in ~ 12 s under `pytest -p no:xdist -m slow tests/test_mms_logc_m JSON + PNG artifacts land in StudyResults/mms_logc_muh_multi_ion_stern/`.

7 Implementation note: H_2O_2 bulk concentration

Production runs use $c_{0,\text{H}_2\text{O}_2}^{\text{bulk}} = \text{H2O2_SEED_NONDIM} = 10^{-4}$, a "seed" value chosen so that the log-concentration formulation can start from a finite $u_{\text{H}_2\text{O}_2}$ at cold-start; the actual $c_{\text{H}_2\text{O}_2}$ at the

electrode grows several orders of magnitude away from this seed via voltage continuation. The MMS test, in contrast, starts Newton directly at $U = U_{\text{ex}}$ with no continuation, so any test-time choice of $c_{0,H_2O_2}^{\text{bulk}}$ must place the manufactured state inside Newton's basin of attraction.

At $c_{0,H_2O_2}^{\text{bulk}} = 10^{-4}$ the H_2O_2 sub-equation is Newton-ill-conditioned for the parallel- $2e/4e$ topology used here. The cathodic R_{4e} has stoichiometry $s_{H_2O_2} = 0$, so the only BV coupling on H_2O_2 is the $+1 \cdot R_{R_{2e}}$ term ($R_{R_{2e}} \approx 2.5$ at $V_{\text{RHE}} = +0.55$ V), and the diagonal block of the Jacobian for H_2O_2 scales as $c_{H_2O_2}^{\text{ex}} \approx 10^{-4}$. Empirically, Newton accepts a step that drives $u_{H_2O_2}$ to ~ -25 in the interior — a state where the H_2O_2 sub-equation is residual-satisfied ($c_{H_2O_2} \rightarrow 0$ collapses every H_2O_2 term) but $u_{H_2O_2}$ has wandered $\mathcal{O}(15)$ away from $u_{H_2O_2}^{\text{ex}}$, giving $\|u_{H_2O_2}\|_{L^2} \sim 20$ instead of the $\sim 10^{-3}$ achieved on every other field.

Setting $c_{0,H_2O_2}^{\text{bulk}} = 1$ for the MMS test problem rebalances the H_2O_2 Jacobian column against the other species; Newton then converges in 2–3 iterations to the manufactured solution. The operator under test is unchanged — $c_{0,H_2O_2}^{\text{bulk}}$ enters only the bulk Dirichlet BC, the bulk packing constant θ_b (contribution $a_{H_2O_2} c_{0,H_2O_2}^{\text{bulk}} = 2.42 \times 10^{-5}$, negligible against $\theta_b \approx 0.991$), and the BV c_{ref} references for the anodic branch — so the convergence rates above verify exactly the production stack the demo script uses, with one unrelated BC value rescaled. The test-suite constant `H2O2_CO_MMS = 1.0` in `scripts/verification/mms_pnpbv_muh_multi_ion_stern.py` carries this note inline.